January 2026

IPC Online

Section: A1

Time: 30 min

Run three threads.

a. Thread1 prints ‘A’ in an infinite loop
b. Thread2 prints ‘B’ in an infinite loop
c. Thread3 prints ‘C’ in an infinite loop

Now, these three threads should run independently. When at least one A, at least one B, and at least one C have been printed in a line, a new line is printed. Continue printing until N lines have been printed.
Input format: <Number of iterations>

Input: N = 3
Output:

BAC	[Line 1]
BABBBAC	[Line 2]
CCCACACACAAAAB	[Line 3]

Note that the letters in a line can appear in any order.

Implement this using semaphore only. [Hint: Only one semaphore lock will suffice for this code.]

Create a file <student_id>.cpp and submit it.